

Commitment To Cooperation With Self-Negotiated Contracts

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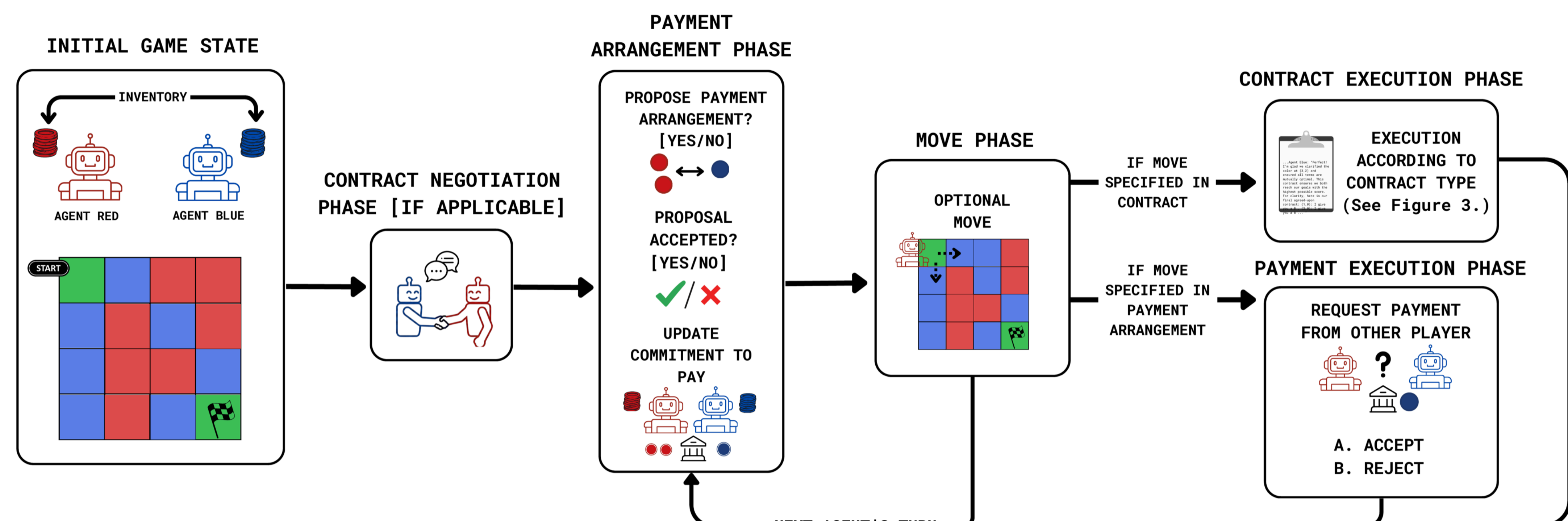
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Why Contracts?

As increasingly capable AI agents interact in multi-agent systems and swarms, agent-to-agent alignment and reliable cooperation become increasingly important. Cooperation is difficult when its costs come early and its benefits later, so each agent has an incentive to defect. Contracts can make commitments credible by making agreements observable and enforceable.

Can LLM agents use self-negotiated contracts to cooperate with commitment?

The CT-Bench Environment



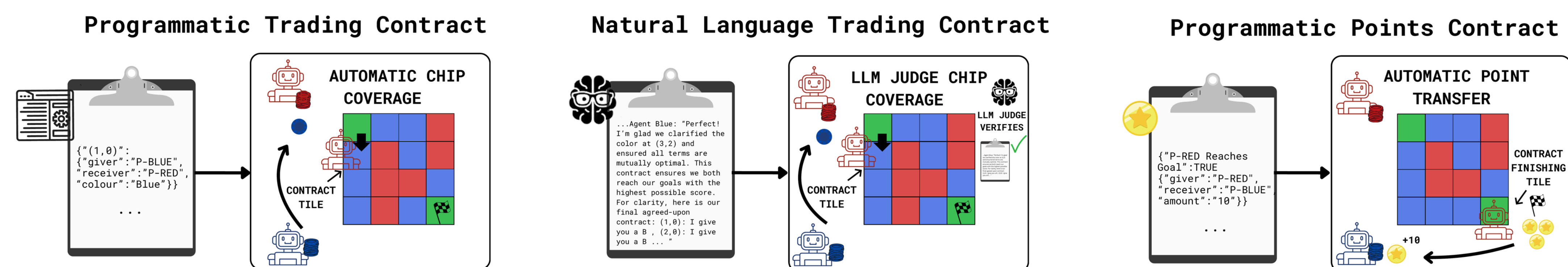
One game: negotiate, optionally form a contract, then move and pay.

P-Red and **P-Blue** hold chips of only their own color, so finishing requires trades; the goal is worth 20 points plus 5 per chip. Pay-for-Partner promises are **non-binding**: a partner can renege.

Setup

Boards	80: 20 Independent (finish alone), 20 Mutually Dependent (must trade), 40 Asymmetric (P-Red can finish alone, P-Blue cannot)
Models	GPT-4.1, Haiku-4.5, LLaMA Maverick, LLaMA Scout, Qwen-3-235B, Qwen-3-30B; self-play
Metrics	Normalized Joint Reward, Gini, Both Finished, Both Beat Baseline (BBB), Defection

Self-Negotiated Contracts



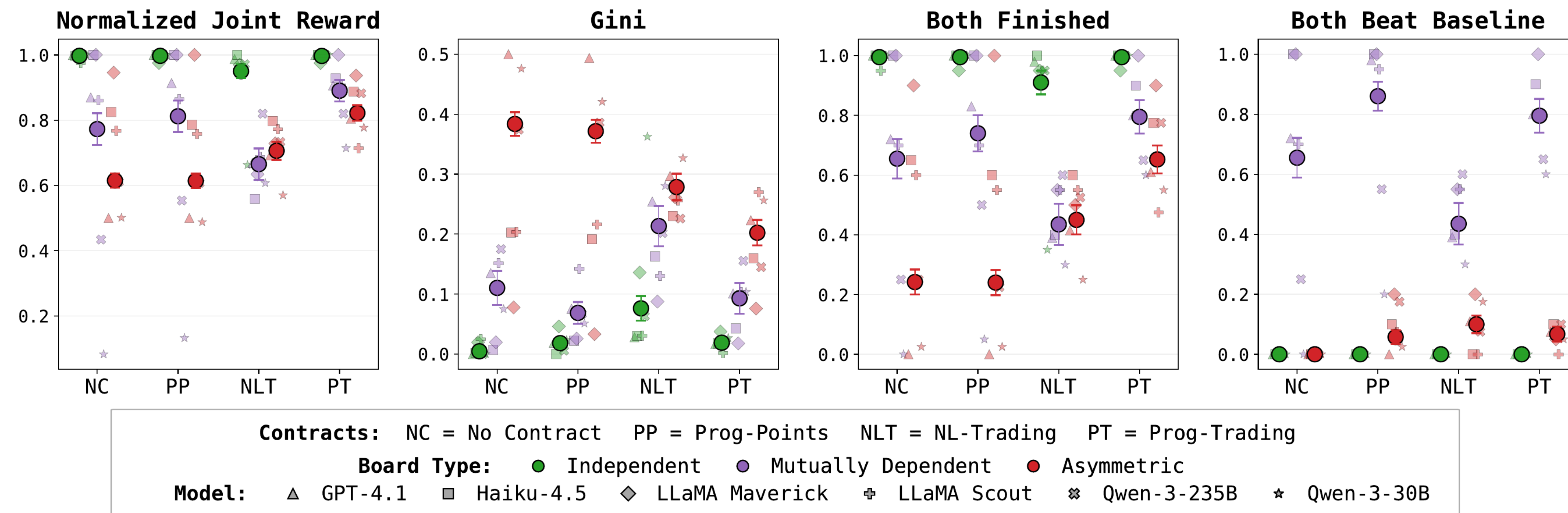
Agents negotiate their own contracts:

- **Programmatic Trading:** chip transfers to cover agreed tiles, auto-executed
- **Natural Language Trading:** contract in natural language, an LLM judge enforces it
- **Programmatic Points:** promise part of your final score, auto-paid

A judge model (Qwen-3-235B) writes the contract; both players must accept. Acceptance: 86% (Haiku-4.5: 67%).

Contracts Improve Cooperative Outcomes

Without contracts, no model ever got both players above baseline on any Asymmetric board (BBB = 0.00). With contracts:



Programmatic contracts improve cooperative outcomes.

Normalized Joint Reward 0.77 → 0.89 under Programmatic Trading (p<0.001).

The gains are consistent across dependent boards.

Highest Both Finished on 45 of 60 dependent boards, vs 6 for No-Contract.

Enforced contracts reduce reliance on non-binding promises.

38% fewer Pay-for-Partner arrangements.

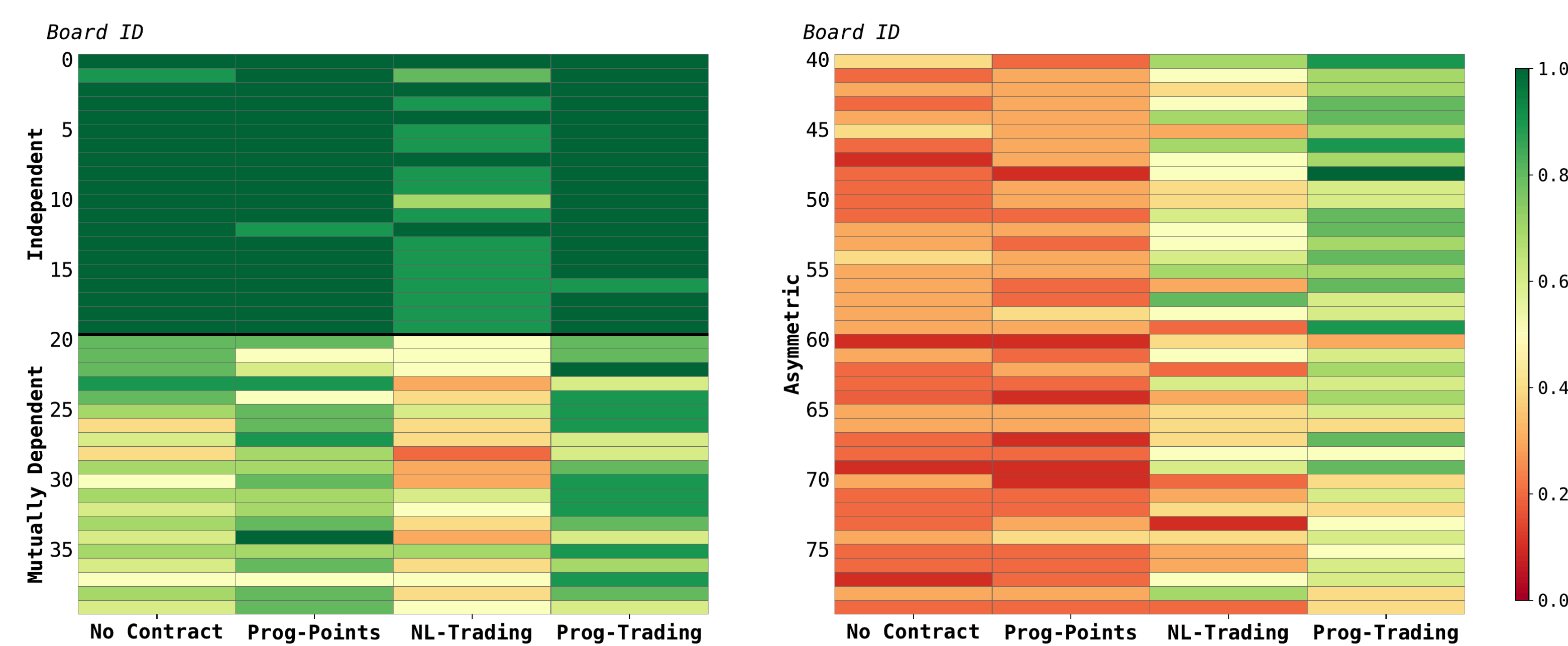
On Asymmetric boards, the gains go to the weaker player.

P-Blue earns 68% more under Programmatic Trading (p<0.001); **P-Red** does not gain.

The form of the contract shapes how fair the deal is.

Programmatic Points: winner-takes-all on 39% of Asymmetric boards; Programmatic Trading: equal tiles in 49%.

Where Contracts Help: Board-Level Outcomes

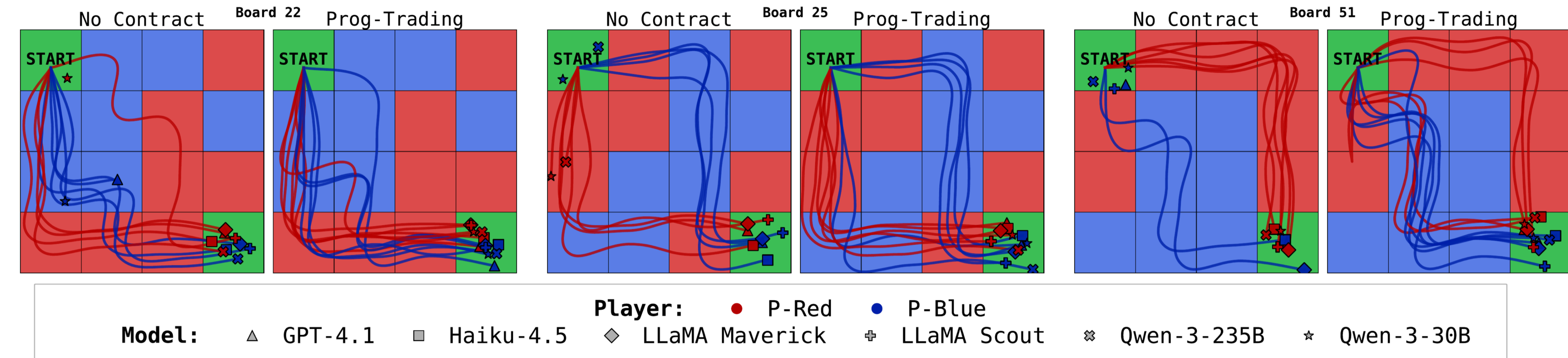


Mean Both Finished per board. Gains hold across most boards; Independent boards succeed in every condition; Asymmetric boards fail most without enforced contracts.

Future Work

- **Beyond self-play:** heterogeneous pairings, misaligned objectives, human-agent settings (in mixed pairs, Programmatic Trading enforced transfers even when Qwen-3-30B honored 6% of promises)
- **Richer environments:** larger boards, incomplete information, more than two players (collusion)
- **Repeated games:** trust, reputation, and defection over time
- **Richer contract mechanisms:** enforcement variations, formation costs, penalties for defection, renegotiation

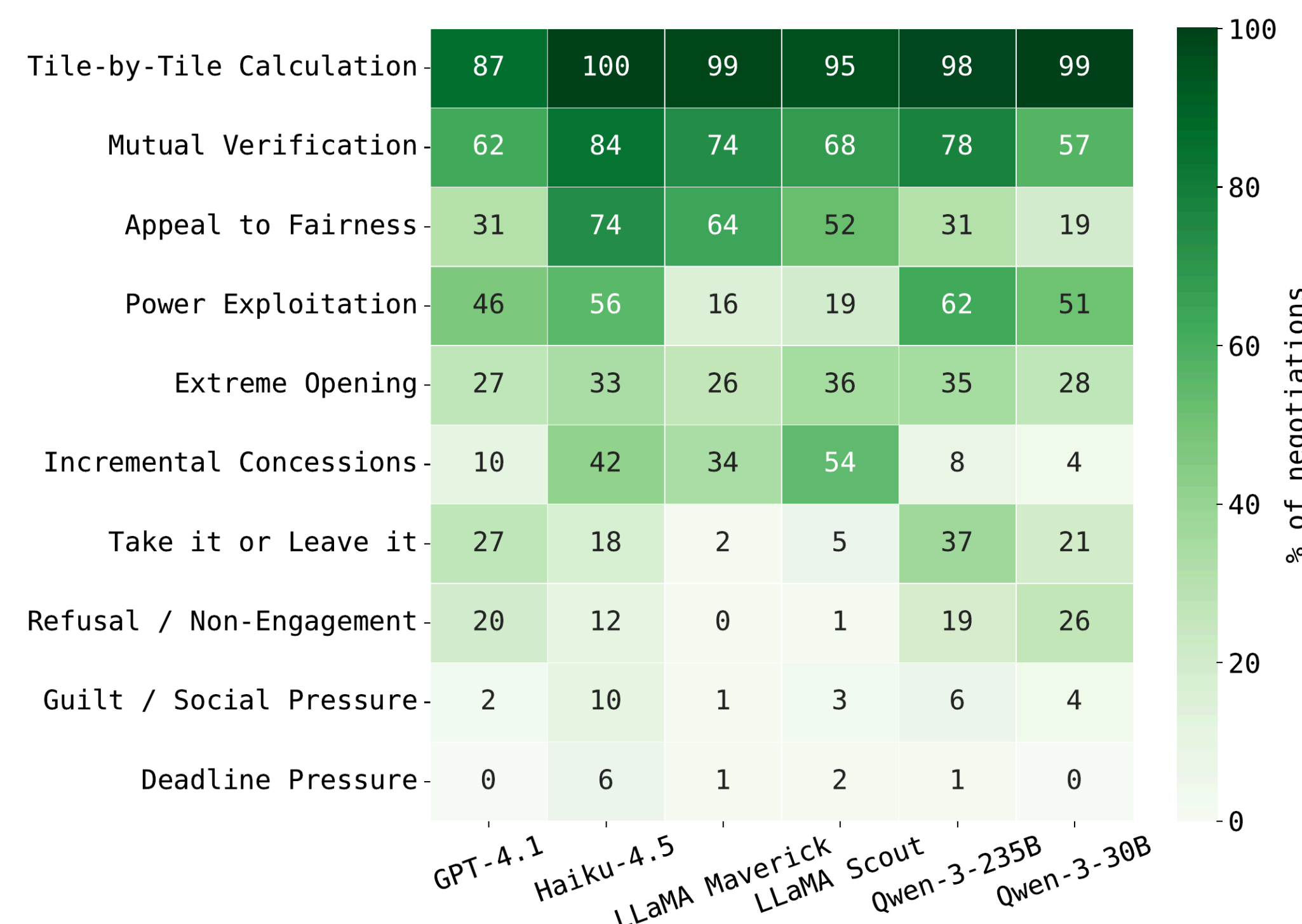
Contracts Change Behavior on the Board



Contracts can improve outcomes. On boards where models fail to reach their goals without a contract, adding a Programmatic Trading contract lets every model complete its route.

But contracts can also backfire. On Mutually Dependent boards, natural-language contracts do worse than no contract: Both Finished 46% vs 61% (none) and 79% (Programmatic Trading). Models over-anchor: they treat the contract as enough, trade less (1.4 vs 3.0 proposals per game), and miss trades they still need.

Negotiation Tactics



Tactic prevalence by model. Haiku-4.5 appeals to fairness most (74% of contracts vs 39% for other models); LLaMA models use power exploitation least (17.5% vs 54%).

Takeaways

1. **The way contracts are represented and enforced matters:** contracts that compile to programs beat natural-language ones (result panels; route maps)
2. **Self-negotiated contracts can improve cooperative outcomes** beyond regular trading (Joint Reward and Both Finished panels)
3. **Agents struggle to reach mutually beneficial outcomes from asymmetric starting points** (BBB panel; heatmap)

Paper & Acknowledgments



CT-Bench is open-sourced.

Paper: arxiv.org/abs/2607.22750

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